

Implant offset layer creation

Scripts

This repository provides scripts for creating concentric layers in a rectangular implant section for image analysis in QuPath or Arivis.

Arivis

build_implant_layers_v1.py This script was built for Arivis Pro 4.4, and is adapted from the matryoshka script by Maurizio Abbate from Zeiss.

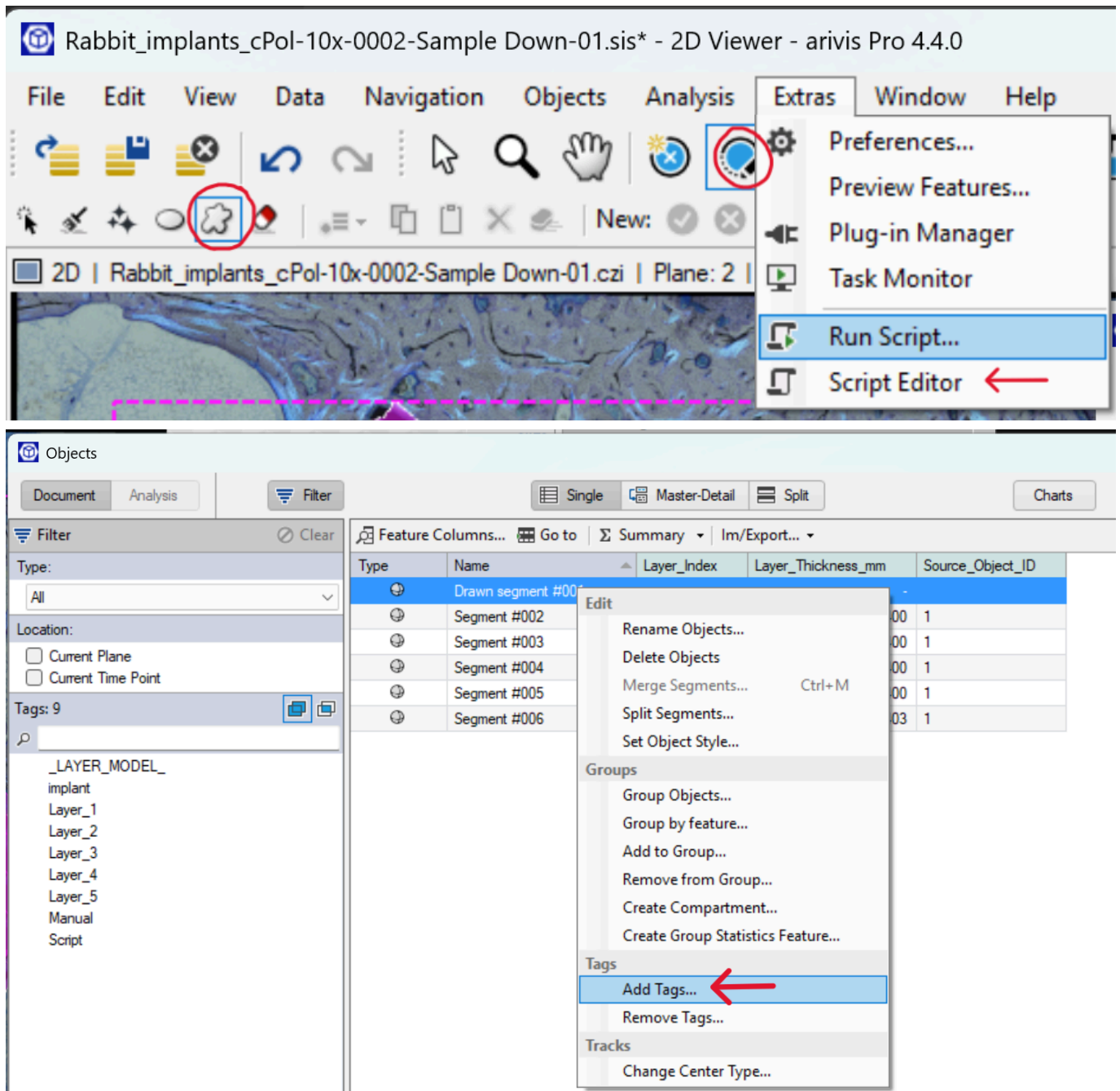
The script can be run from Arivis by going to the menu Extras - Script Editor. From there, the script can be opened by clicking on File - Open, and run using the green Run button.

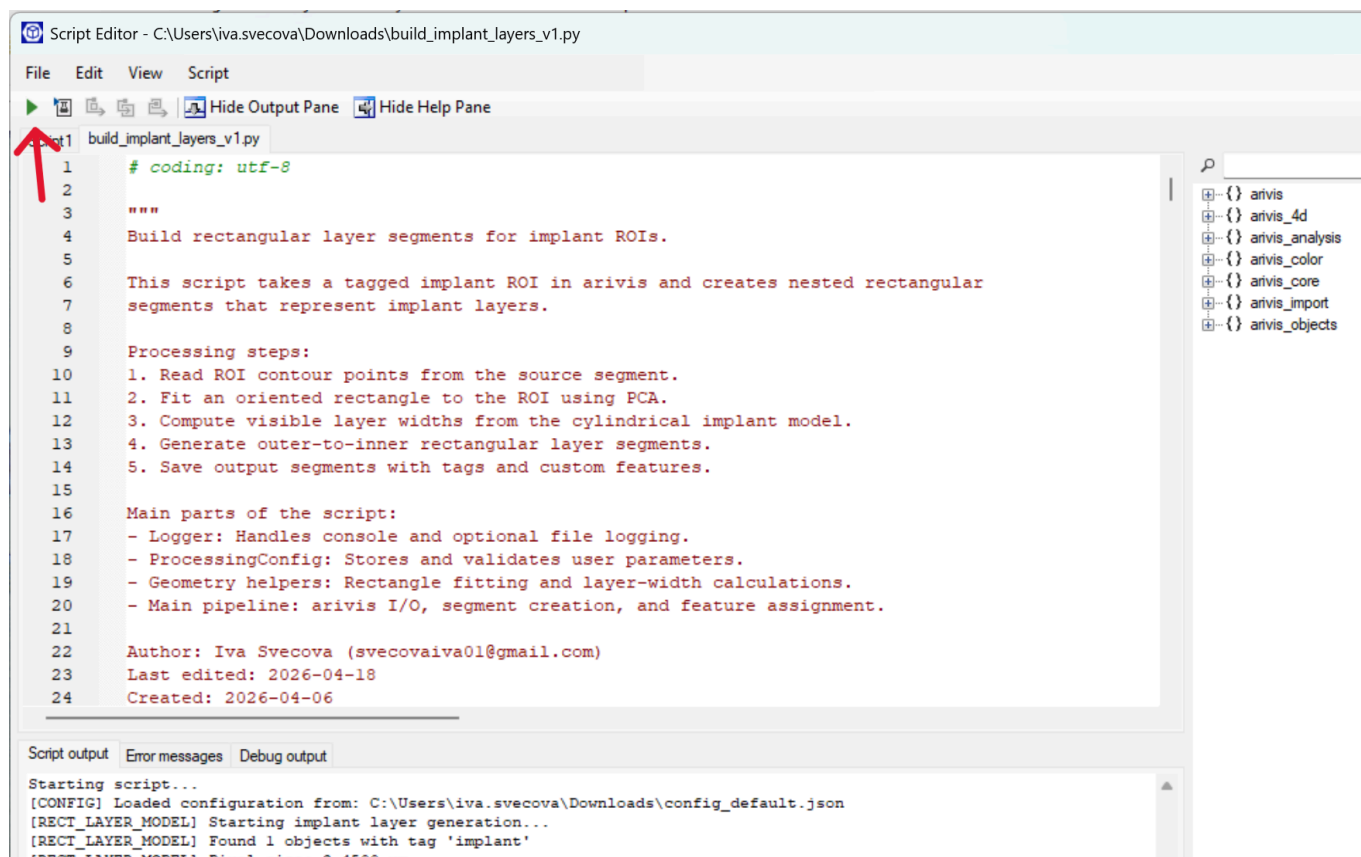
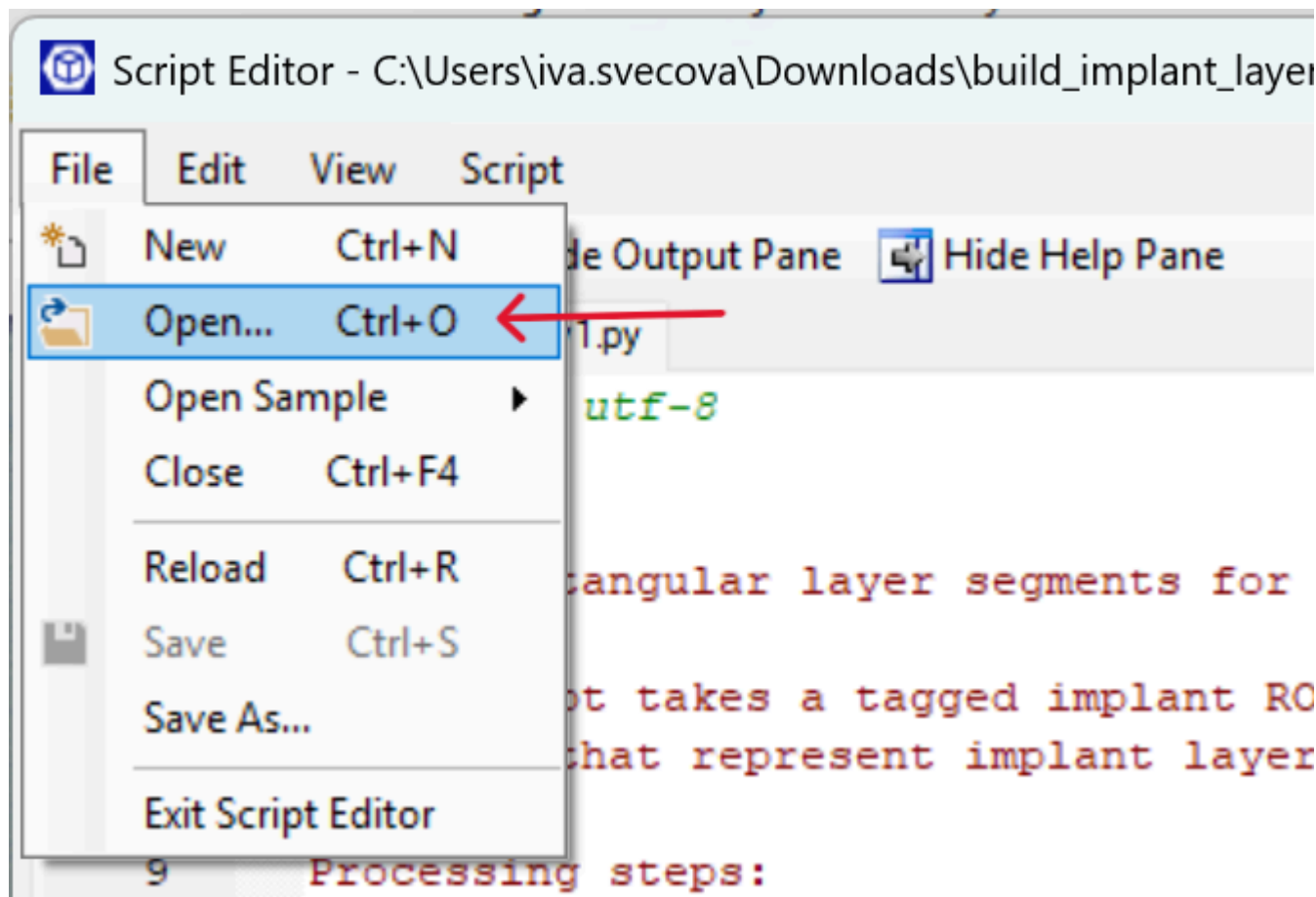
The parameters of the implant and the folder where logs will be stored can be edited in the config file:

config_default.json

This file has to be saved in the same folder as the Python script.

The script requires a hand-drawn object with an assigned tag (by default 'implant', can be changed in the config file). It turns this ROI into a rotated rectangle and divides it into layers.





QuPath

There are two alternative workflows:

1. Automatic workflow

- **01_detectAndMakeLayers.groovy**

Automatically detects a dark rectangular implant in the image, fits a rectangle using thresholding and PCA, and generates concentric rectangular layers as annotation objects.

- **Input:** Image with a single, dark rectangular implant area.
- **Output:** Concentric rectangular layers with thickness measurements in mm.

Alternative:

- **01b_detectAndMakeLayers_withOuterLayer.groovy**

Same as [01_detectAndMakeLayers.groovy](#), but additionally calculates an outer layer with a user-defined width (default 0.5 mm).

2. Manual workflow

If automatic detection is not suitable, use the manual path:

- **M01_polygonToRectangle.groovy**

Converts a user-drawn polygonal ROI (around the area of interest) into a fitted rotated rectangle with a fixed long side (implant height).

- **Input:** Selected polygonal annotation.
- **Output:** Rotated rectangle annotation matching the implant section.

- **M02_makeLayers.groovy**

Takes the fitted rectangle (from M01) and creates concentric rectangular layers, calculating thickness based on a cylindrical model.

- **Input:** Selected rectangle annotation.
- **Output:** Concentric rectangular layers with thickness measurements in mm.

Alternative:

- **M02b_makeLayers_withOuterLayer.groovy**

Same as [M02_makeLayers.groovy](#), but additionally calculates an outer layer with a user-defined width (default 0.5 mm).

- **M03_addOuterLayer.groovy**

Post-hoc alternative to [01b_detectAndMakeLayers_withOuterLayer.groovy](#) or [M02b_makeLayers_withOuterLayer.groovy](#). Use this script when the concentric layers have already been generated without the outer expanded layer, and the outer layer should be added afterwards.

- **Input:** Selected Layer 1 annotation, expected to represent the current outermost implant layer.
- **Output:** An intermediate implant rectangle annotation and a new "Outer layer" annotation with a user-defined width.

Manual path steps:

1. Draw a polygon ROI around the implant area.
2. Run **M01_polygonToRectangle.groovy** to fit a rectangle.
3. Run **M02_makeLayers.groovy** to generate layers (or use M02b for adding also the outer layer).

4. If layers were generated without the outer expanded layer, select Layer 1 and run **M03_addOuterLayer.groovy** to add it post-hoc.

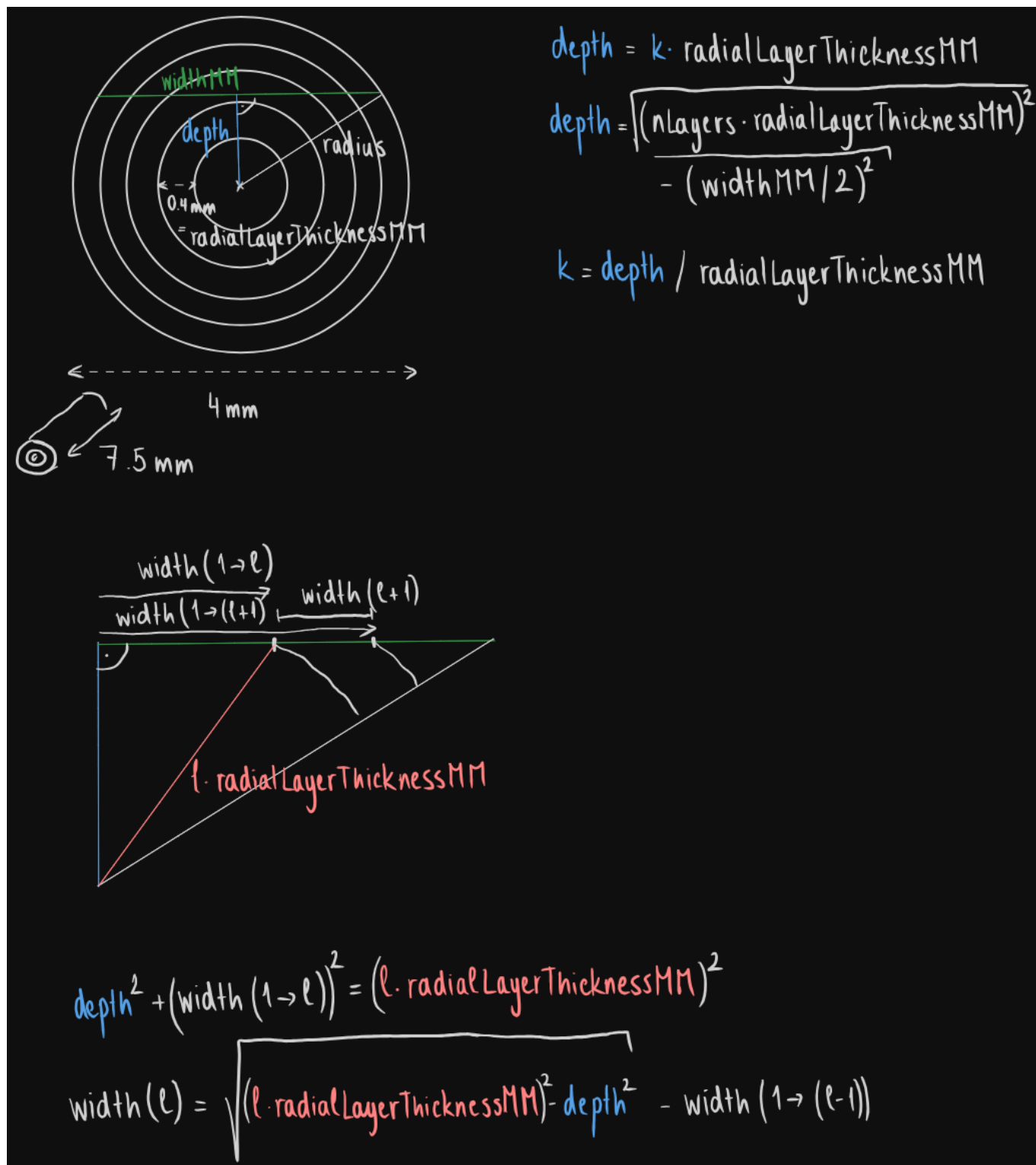
How to use

The groovy scripts are meant to be [ran from QuPath](#).

Rationale behind layer thickness calculation

The script assumes that the region of interest comes from cutting a cylinder perpendicular and offset to the main axis.

[Offset_layer_thickness_computation.ipynb](#) contains calculation of the thickness based on the trigonometrical model shown on the attached figure:



Author and acknowledgement

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If you use the QuPath script for a publication, please cite QuPath: Bankhead, P. et al. QuPath: Open source software for digital pathology image analysis. Scientific Reports (2017). <https://doi.org/10.1038/s41598-017-17204-5>

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Time breakdown

Activity	Estimated duration (hours)	Date
Initial automated implant detection design	1.5	
Segmentation and layer generation in QuPath	3	
Mathematical description of the layers cut off-centre	2	
Implementation of the mathematical model in QuPath	2	
Examination of Arivis-Python interface	3	
Adjusting Arivis-Python script to fit the mathematical model	4	
Documentation, troubleshooting	2	
Meetings, discussions	2	
Update to enable flexible implant length and selection of edge to segment into layers	1	2026-08-04
Adding the optional outer layer during layer generation	1	2026-09-02
Adding the M03 script to add the outer layer post-hoc	3	2026-09-07, 2026-09-16